

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – FINAL EXPLANATION: WHAT CAUSES THE DIFFERENCES IN GERMINATION, EMERGENCE AND GROWTH OF MAIZE AND BEAN PLANTS?

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Your task is to write a Claim-Evidence-Reasoning (CER) explanation that answers the driving question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Think back to the farmer who planted maize and beans on the same day after the rain, yet some seeds sprouted quickly, some were slow, and some healthy-looking seeds did not germinate at all. Use evidence from our unit: seed structure observations, germination investigations, dormancy investigations, observations of primary and secondary growth, and evidence on plant hormones. Write in your own words, use correct scientific terms, and refer to your own data. Finally, revise your Day 1 model or drawing to show how your thinking has changed. The exemplars below show what a strong answer looks like. Use them as a guide, not something to copy.

Section 1: Claim

Write a clear claim that answers the driving question. Name the main causes of the differences you saw in the farmer's field (seed structure, conditions for germination, dormancy, and the processes and hormones that control growth). Do not give details yet. Just state your answer in 2-3 sentences.

The differences in germination, emergence and growth of maize and bean plants are caused by several factors working together. These are the different structures of maize and bean seeds, whether the seeds get the conditions they need (water, oxygen and a suitable temperature), whether a seed is dormant, and the way cell division, cell enlargement and plant hormones control growth. Growth is also more than getting bigger, because the plant changes its structure and function as it develops from a seed into a mature plant.

Section 2: Evidence from Germination, Dormancy and Seed Structure

Describe at least three pieces of evidence from our investigations. Include (a) what

In our germination investigation, maize and bean seeds that had water, air and a warm temperature germinated and grew radicles and shoots. Seeds that were kept dry did not germinate, seeds that were submerged in water with little oxygen did not germinate normally, and seeds kept in a very cold place stayed unchanged. This shows that each condition is needed. When we dissected

happened to seeds given water, oxygen and a suitable temperature compared with seeds missing one condition, (b) what you observed about the structure of maize (one cotyledon) and bean (two cotyledons) seeds and how their seedlings emerged, and (c) anything you learned about dormancy in seeds that looked healthy. Use your own observations or data, such as how many seeds germinated, how many days it took, and what you drew.	the seeds, the bean seed had two large cotyledons holding stored food, while the maize grain had one cotyledon and a starchy endosperm. When they germinated, the maize and bean seedlings emerged and developed in different patterns. In the dormancy investigation, some seeds that looked healthy did not germinate until the dormancy factor (for example a hard seed coat or a need for a certain condition) was dealt with. This explains why in the farmer's field some healthy seeds did not sprout even though the rain was the same.
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Section 3: Evidence from Growth and Plant Hormones

Describe the evidence you gathered about how plants grow after germination. Include what you learned about primary growth (increase in length of roots and shoots), secondary growth (increase in thickness of stems and roots in woody plants such as trees), and how hormones such as auxins affect growth, for example shoots bending towards light. Mention at least two observations or results.	When we measured growing seedlings, the roots and shoots became longer over time. Slides and diagrams showed that cell division in the meristems at the tips and cell enlargement behind them cause this increase in length, which is primary growth. We also saw that in woody plants, lateral meristems add new tissue so that stems and roots become thicker. This is secondary growth. That helps explain why the farmer saw stems becoming taller and thicker, and roots going deeper. In the light experiment, seedlings bent towards the light source. This is because auxins cause cells on the shaded side to elongate more, so the shoot curves towards the light. This showed that hormones control the direction and amount of growth.
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Section 4: Reasoning

Explain HOW and WHY your evidence supports your claim. Link each piece of evidence to a scientific idea: water activates enzymes and metabolism; stored food gives energy and materials; dormancy delays germination; monocot and dicot	Water enters the seed and activates enzymes, which start the metabolic processes that let the embryo grow. Oxygen is needed for respiration to release energy, and a suitable temperature lets enzymes work well. The stored food in the cotyledons or endosperm supplies energy and materials until the seedling can make its own food. That is why seeds lacking one condition did not germinate normally. Some seeds stayed dormant because internal or external factors blocked germination until the right conditions came or dormancy was broken, so seeds planted on the same day did not all sprout together. Maize is a monocot and bean is a dicot, so their structures lead to different ways of emerging and developing. After emergence, cell division and cell enlargement increase length (primary growth), lateral meristems increase thickness (secondary growth), and auxins control elongation and direction. So the differences the farmer saw come from seed structure, conditions, dormancy and growth control together. Growth and
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structures affect emergence; cell division and enlargement cause growth; hormones regulate growth. Finish by explaining why this matters for farmers in Kenya.	development let plants produce leaves, roots, flowers, fruits and seeds. For farmers in places like the Rift Valley, knowing this helps in choosing planting time, preparing seed and managing moisture, which supports maize and bean harvests, food production and food security.
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Section 5: Revise Your Day 1 Model

Look at your Day 1 model or drawing of why the maize and beans behaved differently. Draw a new model that includes seed structure, the conditions for germination, dormancy, primary and secondary growth, and the role of hormones such as auxins. Label it. Then write 3-4 sentences describing what you added, changed or corrected, and why.	On Day 1, my drawing only showed rain falling on seeds and plants growing, and I thought the seeds that did not germinate were simply bad. In my new model I drew the inside of a maize grain (one cotyledon, endosperm, embryo) and a bean seed (two cotyledons, embryo), with arrows showing water, oxygen and warmth entering. I added a label for dormancy to explain healthy seeds that did not sprout. I drew the seedling with growing tips where cells divide and enlarge, a thickening stem for secondary growth, and an arrow showing auxin moving to the shaded side so the shoot bends towards light. My thinking changed because I now know that the differences come from the seed, its conditions and the control of growth, not from the rain alone.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy	All ideas about germination conditions, dormancy, monocot and dicot seeds, primary and secondary growth and auxins are correct and use scientific terms precisely.	Most ideas are correct, with minor errors or some terms used loosely.	Several ideas are incorrect or missing, or terms are used wrongly.
Use of evidence	Uses specific evidence from at least three investigations (such as germination, seed structure, dormancy, growth and hormones) with data or observations.	Uses evidence from two investigations, with some detail.	Uses little or general evidence, or evidence is not from our investigations.
Reasoning	Clearly explains how and why each piece of evidence supports the claim, linking it to scientific ideas such as enzyme activation, stored food, cell division and enlargement, and hormones.	Links most evidence to the claim, but some links are incomplete.	Reasoning is missing or only repeats the evidence without explaining.
Connection to the driving question	Directly answers the driving question,	Answers the driving question and revises	Only partly answers the driving question,

and revised model	refers to the farmer's field, and revises the Day 1 model with clear labels and an explanation of how thinking changed.	the model, but links to the phenomenon or the explanation of change are limited.	and the model is unrevised or unclear.
Clarity of communication	Well organised in claim, evidence and reasoning, easy to follow, with correct spelling and clear, well-labelled drawings.	Mostly organised and clear, with some errors that do not block understanding.	Disorganised or hard to follow, with unclear drawings or labels.

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